

Thermal Safety Analysis of Epiretinal Stimulation

Joule Heating vs. Natural Retinal Metabolism

A COMSOL Multiphysics FEM Study Inspired by the Argus II Retinal Prosthesis

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Modelling of Multi-Physics Phenomena

Project Overview

What

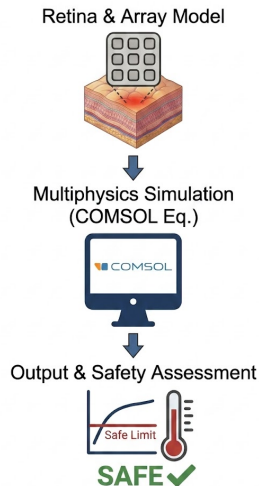
FEM simulation of epiretinal stimulation with a 3×3 Pt electrode array on a multi-layer retinal model

Why

Assess thermal safety by comparing **Joule heating** to the **natural metabolic heat** of the retina

How

COMSOL Multiphysics model integrating the Pennes bioheat equation with layer-specific metabolic heat sources from literature



Clinical Motivation — Why Retinal Prostheses?

- **WHO (2022):** ~43 million blind, ~295 million with moderate-to-severe visual impairment worldwide
 - **Key pathologies:**
 - **Retinitis Pigmentosa (RP):** hereditary photoreceptor degeneration (~1:4000)
 - **Age-related Macular Degeneration (AMD):** leading cause of blindness in developed countries
 - Both diseases destroy **photoreceptors** but leave **retinal ganglion cells (RGCs)** structurally intact
- ⇒ Direct electrical stimulation of surviving RGCs is possible

Healthy Retina

RGC

INL

ONL

✓ Photoreceptors OK

RP / AMD

RGC

INL

ONL

× Photoreceptors lost



Stimulate RGCs
directly

The Thermal Safety Problem

Central Question

How does the **Joule heating** from the electrode compare to the **natural metabolic heat** of the retina itself?

Thermal damage threshold:

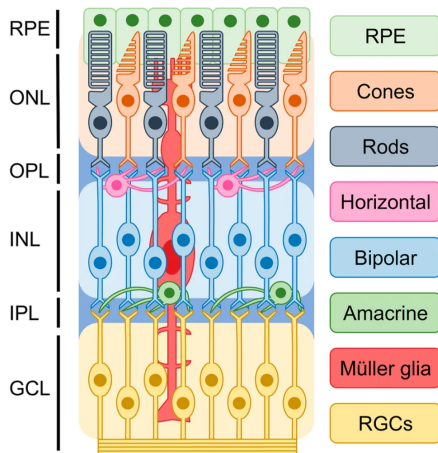
- ISO 14708: $\Delta T < 1^{\circ}\text{C}$ – 2°C for chronic exposure
- Retinal neurons are heat-sensitive

Our approach (5 steps):

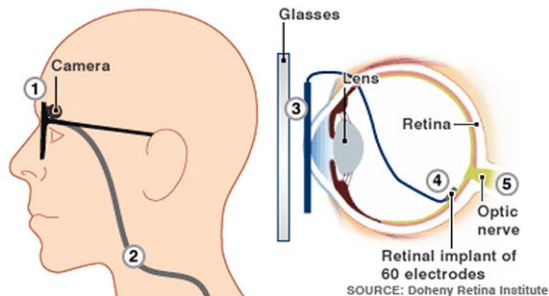
- ① Multi-layer retinal geometry
- ② Metabolic heat from O_2 consumption data
- ③ Biphasic pulse simulation
- ④ Compare ΔT_{Joule} vs. ΔT_{met}
- ⑤ Polyimide insulation effect

Retinal Anatomy — Multi-Layer Structure

Layer	Abbr.	Thickness
Vitreous	VIT	25 μm
Ganglion Cells	RGC	22 μm
Inner Plexif.	IPL	23 μm
Inner Nuclear	INL	27 μm
Outer Plexif.	OPL	16 μm
Outer Nuclear	ONL	31 μm
Subretinal Sp.	SRS	40 μm
Pigment Epith.	RPE	20 μm
Choroid	CHO	426 μm



The Argus II Epiretinal Prosthesis



Parameter	Value	Source
Material	Platinum	[1]
Diameter	200 μm	[1]
Thickness	10 μm	[1]
Pitch	575 μm	[1]
Substrate	Polyimide, 15 μm	[1]
Current	20 μA	[2]
Frequency	50 Hz	Design

[1] Werginz et al., 2015 [2] Biswas et al., 2017

Our simplification: 3×3 electrode grid with real Argus II dimensions

Epiretinal vs. Subretinal Approach

Epiretinal (Argus II)

- Electrodes on **inner** retinal surface
- Stimulates RGCs directly
- Complete bypass of phototransduction
- Simpler surgical fixation

Subretinal (Alpha IMS/AMS)

- Electrodes in subretinal space
- Stimulates bipolar cells
- Uses residual neural circuits
- More complex implantation

Epiretinal challenge: current must traverse intermediate layers $\Rightarrow$ dispersion & Joule heating
 $\Rightarrow$ *This is precisely the thermal effect our project quantifies.*

Oxygen Consumption → Metabolic Heat

Conversion formula:

$$Q_{\text{met}} = \frac{\dot{V}_{\text{O}_2} \times 20 \text{ kJ/L}}{V_{\text{tissue}}}$$

Species	Outer Retina QO_2 (dark-adapted)	Source
Cat	3.9 mL O_2 /(100g·min)	[4]
Rat	1.96 mL O_2 /(100g·min)	[5]
Macaque	4.6 mL O_2 /(100g·min)	[6]

Example: ONL (dark)

$$QO_2 = 4.0 \text{ mL}/(100\text{g}\cdot\text{min})$$

$$\text{Heat} = 4.0 \times 20 = 80 \text{ J}$$

$$\text{Power} = 80/60 = 1.33 \text{ W}$$

$$Q_{\text{met}} = 1.33/10^{-4} = 13\,300 \text{ W/m}^3$$

[4] Braun & Linsenmeier, 1995 [5] Yu & Cringle, 2002 [6] Birol et al., 2007

Metabolic Heat Parameters in the Model

Layer	Q_{met} (W/m ³)	Justification
ONL (Photoreceptors)	10,000	Average cat/macaque, dark-adapted
RPE	4,000	Active transport, phagocytosis
INL	1,500	Net effective heat [*]
RGC	1,500	Net effective heat [*]
IPL	800	Synaptic activity
OPL	800	Synaptic activity
VIT, SRS, CHO	0	Extracellular fluids / perfusion sink

** Inner retina values are “net effective”: the model does not include capillary perfusion, so reduced values compensate for the absence of the vascular cooling term.*

Governing Equations

Pennes Bioheat Equation (1948)

$$\rho C_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \underbrace{\rho_b C_{p,b} \omega_b (T_b - T)}_{\text{Blood perfusion}} + \underbrace{Q_{\text{met}}}_{\text{Metabolic}} + \underbrace{Q_{\text{Joule}}}_{\text{Electrical}}$$

Electric Currents (quasi-static)

$$\nabla \cdot (\sigma \nabla V) = 0$$

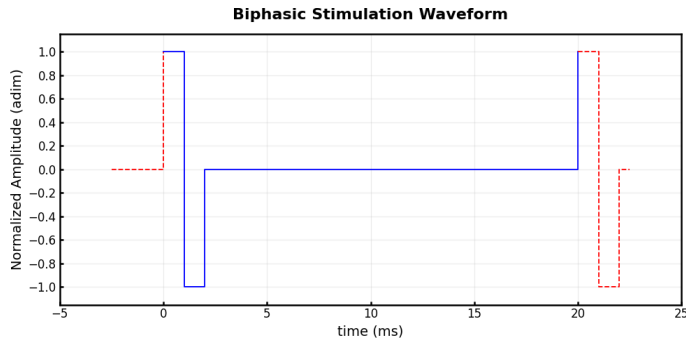
Joule Heating Coupling

$$Q_{\text{Joule}} = \sigma |\nabla V|^2$$

Boundary conditions:

- $T = T_{\text{body}} = 310.15 \text{ K } (37^\circ\text{C})$ at choroid base
- Perfusion: $\omega_b = 0.004/\text{s}$, applied only to the choroid

Biphasic Stimulation Waveform



Parameters:

- $I_s = 20 \mu\text{A}$
- $f = 50 \text{ Hz}$ ($T = 20 \text{ ms}$)
- Cathodic: 1 ms
- Gap: 0.1 ms
- Anodic: 1 ms
- Duty cycle: **10.5%**

Charge-balanced waveform prevents electrochemical damage.
Periodic extrapolation in COMSOL.

3D Model Geometry

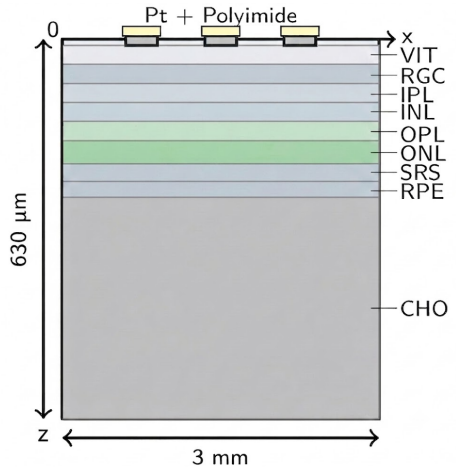
Computational domain:

- 3×3 mm tissue block (XY plane)
- 9 retinal layers stacked along Z
- Total thickness: $\sim 630 \mu\text{m}$

Electrode array:

- 3×3 Pt cylinders ($\varnothing 200 \mu\text{m}$, $h 10 \mu\text{m}$)
- Pitch: $575 \mu\text{m}$ center-to-center
- Polyimide substrate ($15 \mu\text{m}$) above each electrode

Built in COMSOL Multiphysics 6.4



Material Properties

Layer	σ (S/m)	ε_r	k (W/m·K)	Source
Vitreous	1.25	98	0.578	[1]
RGC	0.02262	150,000	0.565	[1]
IPL	0.03717	235,240	0.565	[1]
INL	0.01277	235,240	0.565	[1]
OPL	0.03401	235,240	0.565	[1]
ONL	0.01244	235,240	0.565	[1]
SRS	0.04831	235,240	0.565	[1]
RPE	0.01	235,240	0.565	[1]
Choroid	0.087	150,000	0.565	[1]
Platinum	9.43×10^6	1	71.6	[1]
Polyimide	10^{-14}	3.4	0.12	[1]

[1] Werginz et al., 2015 | $\rho = 1000 \text{ kg/m}^3$, $C_p = 4180 \text{ J/(kg} \cdot \text{K)}$ (tissue)

Physics Interfaces & Study Configuration

Three coupled physics:

- 1 **Electric Currents (ec)** — potential & current distribution
- 2 **Heat Transfer in Solids (ht)** — Pennes bioheat with metabolic sources
- 3 **Electromagnetic Heating (emh)** — coupling: $Q_J = \sigma |\nabla V|^2$

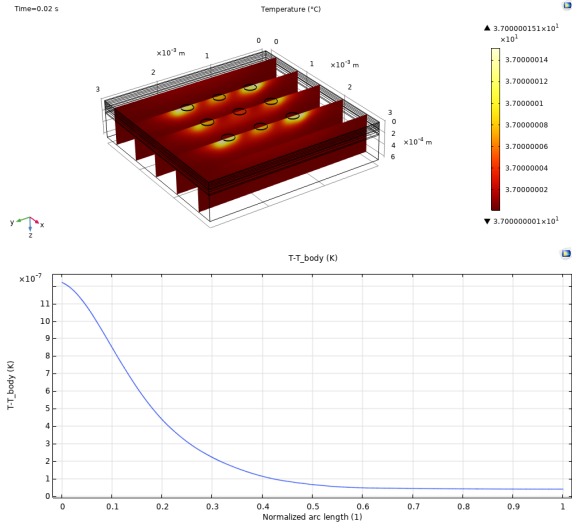
Study	Type	EC	HT	Q_{met}	Poly. sweep
Joule Only	Time Dep. (20 ms)	✓	✓	—	—
Metabolic Only	Stationary	—	✓	✓	—
Combined	Time Dep. (20 ms)	✓	✓	✓	poly_k

Auxiliary Sweep on poly_k (Combined study only): polyimide thermal conductivity toggled to assess insulation effect.

3D Temperature Distribution — Joule Effect

- Joule heating is **highly localized** at the electrode-tissue interface
- Rapid decay with distance from the current source
- Consistent with $Q_J \propto |\mathbf{E}|^2$
- $\Delta T_{\text{max, Joule}} = 8.8824 \times 10^{-6} \text{ K}$

*Slice plot at the end of stimulation time
($t \approx 20 \text{ ms}$)*

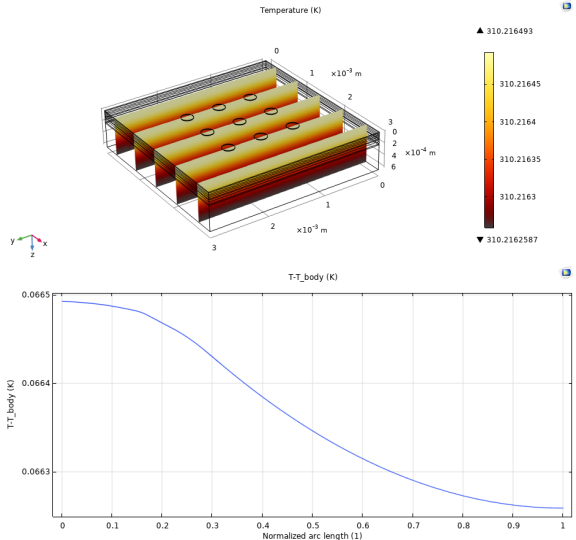


Metabolic Baseline Temperature Profile

- Stationary study: Q_{met} only
- $\Delta T_{\text{max}} \approx 0.066493 \text{ K}$
- Peak located at **ONL** (photoreceptors) — highest metabolic rate
- Consistent with physiological estimates

Key finding

The natural metabolic baseline of the retina produces a thermal signature of $\sim 66 \text{ mK}$.

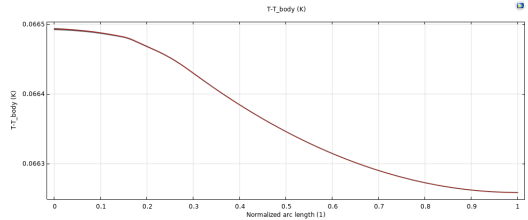
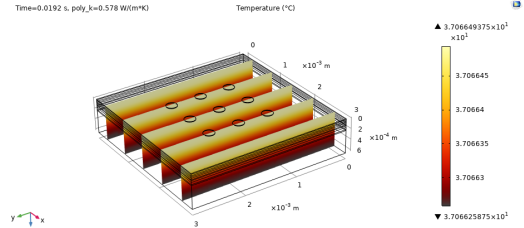


Combined Temperature Distribution — Joule + Metabolic

- Time-dependent study: $Q_{\text{met}} + Q_{\text{Joule}}$ simultaneously
- Includes all heat sources + choroidal perfusion
- $\Delta T_{\text{max, combined}} \approx 0.066\,501\text{ K}$
- Temperature profile dominated by metabolic component

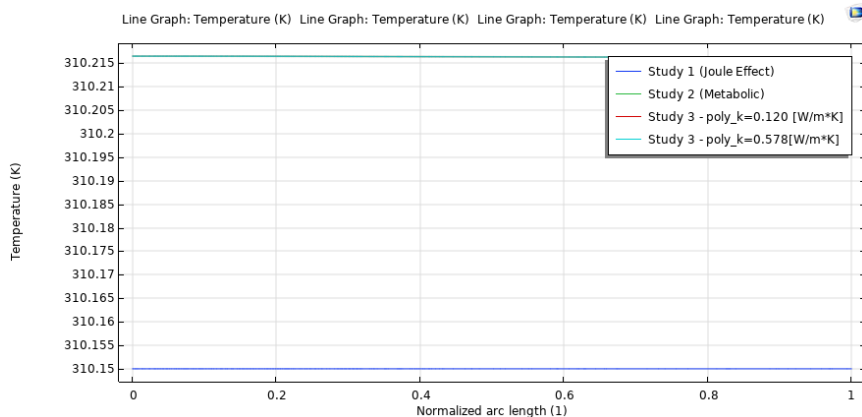
Key observation

The combined ΔT is nearly identical to the metabolic-only scenario, confirming that Joule heating contributes a negligible thermal increment.



Slice plot at the end of stimulation time ($t \approx 19.2\text{ ms}$)

Comparative Temperature Profile $T(z)$



Cut Line 3D along Z, centered under electrode

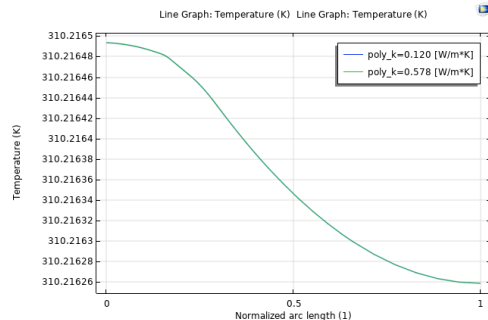
Key result:

- ΔT_{Joule} is **negligible** relative to the metabolic baseline ΔT_{met}
- Linear superposition confirms the system operates in a small- ΔT regime

Effect of Polyimide Thermal Insulation

- Polyimide: $k = 0.12 \text{ W}/(\text{m} \cdot \text{K})$
($\sim 4.7\times$ less conductive than tissue)
- In theory, acts as a unidirectional thermal barrier
- **Auxiliary Sweep result:**

Scenario	$\Delta T_{\max}$
With polyimide ($k = 0.12$)	0.066 501 K
Without polyimide ($k = 0.578$)	0.066 500 K
Difference	$\sim 1 \mu\text{K}$



Conclusion

At $20 \mu\text{A}$, polyimide insulation has **no measurable effect** because $Q_{\text{Joule}} \ll Q_{\text{met}}$ — further confirming thermal safety.

Summary of ΔT values:

Scenario	$\Delta T_{\max}$
Metabolic Only	$\approx 0.066\,493\text{ K}$
Joule Only	$\approx 8.8824 \times 10^{-6}\text{ K}$
Combined	$\approx 0.066\,501\text{ K}$
Safety limit	$<1\text{ }^{\circ}\text{C}-2\text{ }^{\circ}\text{C}$

Model limitations:

- 1 No inner retinal capillary perfusion
- 2 Simplified geometry (3×3 vs. 6×10)
- 3 Uniform, isotropic material properties
- 4 Single-cycle simulation (20 ms)

Conclusions

- ① 3D FEM model accurately reproduces multi-layer retinal structure
- ② Metabolic heat values are consistent with literature (Braun, Yu, Birol)
- ③ Joule heating is 0.013% of the metabolic baseline
- ④ Polyimide insulation has negligible effect ($\Delta T \sim 1 \mu\text{K}$) at 20 μA

Future developments

- Parametric sweep on current amplitude and electrode diameter
- Chronic regime simulation (average power, stationary)
- Inner retinal capillary perfusion (full Pennes model)
- *In vivo* experimental validation

- ① Werginz, P., et al. (2015). Modeling the response of ON and OFF retinal bipolar cells during electric stimulation. *BioMedical Eng. OnLine*, 14(S1), S65.
- ② Biswas, D. K., et al. (2017). Design of electrode configuration for in vitro experiments of epiretinal prosthesis. *IET Sci. Meas. Technol.*
- ③ Loizos, K., et al. (2023). A bidomain model of epiretinal stimulation. *arXiv:2303.04685v1*.
- ④ Braun, R. D., Linsenmeier, R. A., & Goldstick, T. K. (1995). Oxygen consumption in the inner and outer retina of the cat. *IOVS*, 36(3), 542–554.
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- ⑥ Birol, G., et al. (2007). Oxygen distribution and consumption in the macaque retina. *Am. J. Physiol.*, 293, H1696–H1704.
- ⑦ Pennes, H. H. (1948). Analysis of tissue and arterial blood temperatures in the resting human forearm. *J. Appl. Physiol.*, 1(2), 93–122.
- ⑧ Vision Loss Expert Group of the Global Burden of Disease Study, et al. (2024). Global estimates on the number of people blind or visually impaired by cataract: a meta-analysis from 2000 to 2020. *Eye*.

Thank you for your attention

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